

What Drives Variability in Tropical Cyclone Landfall Patterns in the Eastern North Pacific? Environmental Driver and Implications

Jose A. Ocegueda Sanchez^{1*}, Daniel R. Chavas¹ & Jhordanne J. Jones^{2,1}

¹Purdue University, Department of Earth Atmospheric and Planetary Sciences

²UCAR, Cooperative Programs for the Advancement of Earth Science

*jocegue@purdue.edu

Motivation

- The Eastern North Pacific (**ENP**) has the **highest Tropical Cyclone (TC) density** in the world (Jiuwei et al., 2020) but receives relatively little research attention, particularly regarding **landfalls**.
- In **México**, TCs are the natural disasters with the **most significant economic and social impact** on average, affecting **not only during landfall but also** increasing operative expenses **in the following months** (Juárez-Torrez M. & Puigvet J. 2021).
- During **2023**, the **ENP** experienced **two** of its top 10 **costliest TCs**: Hurricane Hilary and Otis (García-Franco et al., 2024).

Research questions

- What **controls** the **landfall variability** in the ENP? Is the variability due to changes in **genesis** or variations in **steering flow**?
- How are the **landfall patterns** in the ENP during years with **different landfall probabilities**?
- Why does the ENP have a **bimodal distribution** in **total landfall counts**?

Framework

We start defining the **landfall probability** (p_{lf}) as the ratio of the number of storms that make landfall (N_{lf}) to the total number of storms that form (N_{gen}) for a given period.

$$p_{lf} \equiv \frac{N_{lf}}{N_{gen}}$$

We analyze **two groups** of years based on their landfall probability: the high (**HLF**) and the low (**LLF**) landfall years.

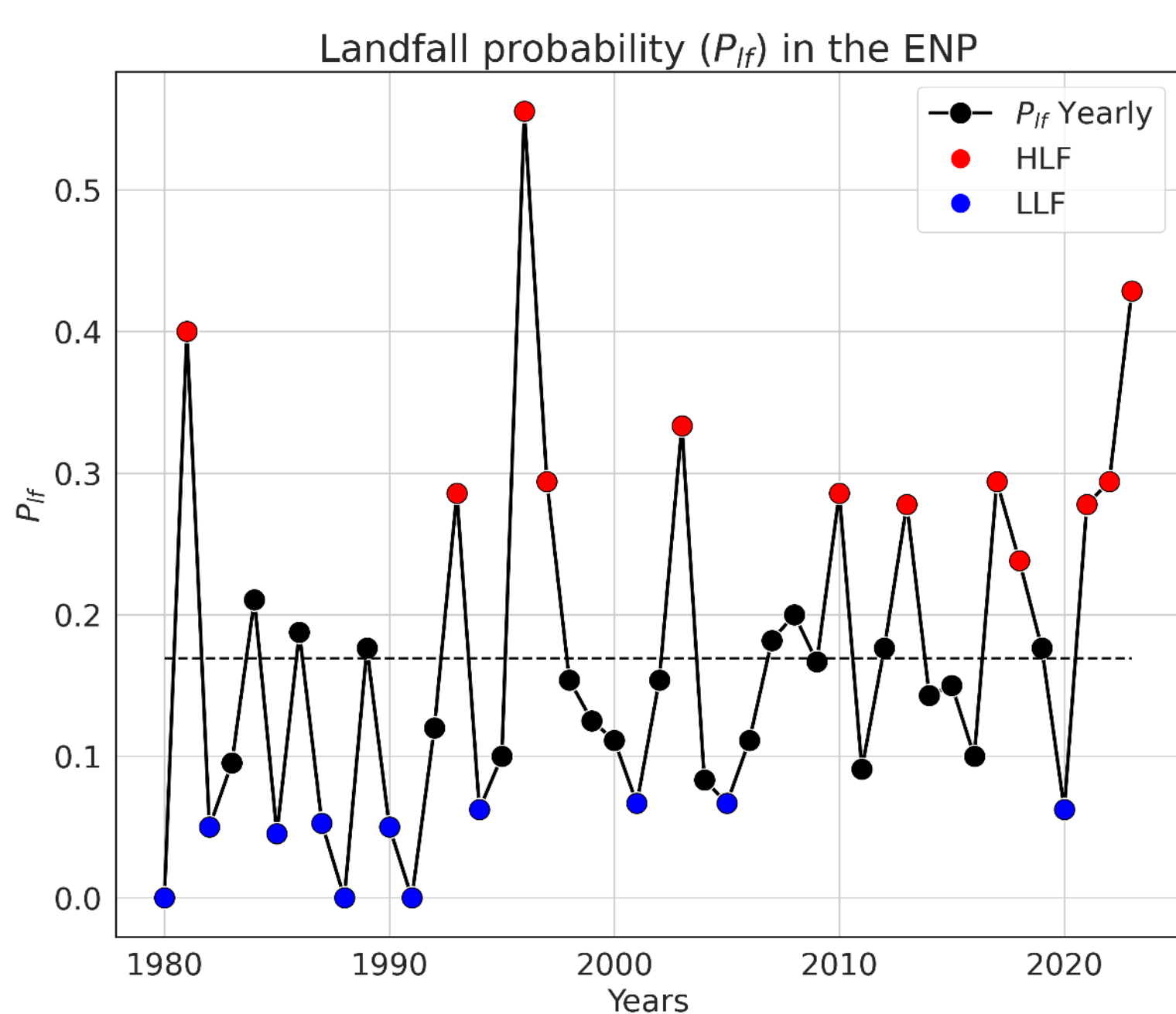


Fig 1. Yearly Landfall Probability in the ENP. The high and low landfall years are years outside the IQR.

Then, we use ERA5 reanalysis data to decompose total landfall counts as follows:

$$N_{T,lf} = \sum_{\phi, \lambda} \underbrace{\overline{p_{lf}} \overline{N_{gen}}}_{\text{Mean/Average landfalls}} + \underbrace{\overline{p'_{lf}} N'_{gen}}_{\text{Landfalls due to genesis anomalies}} + \underbrace{p'_{lf} \overline{N_{gen}}}_{\text{Landfalls due to probabilities anomalies}} + \underbrace{p'_{lf} N'_{gen}}_{\text{Residual term}}$$

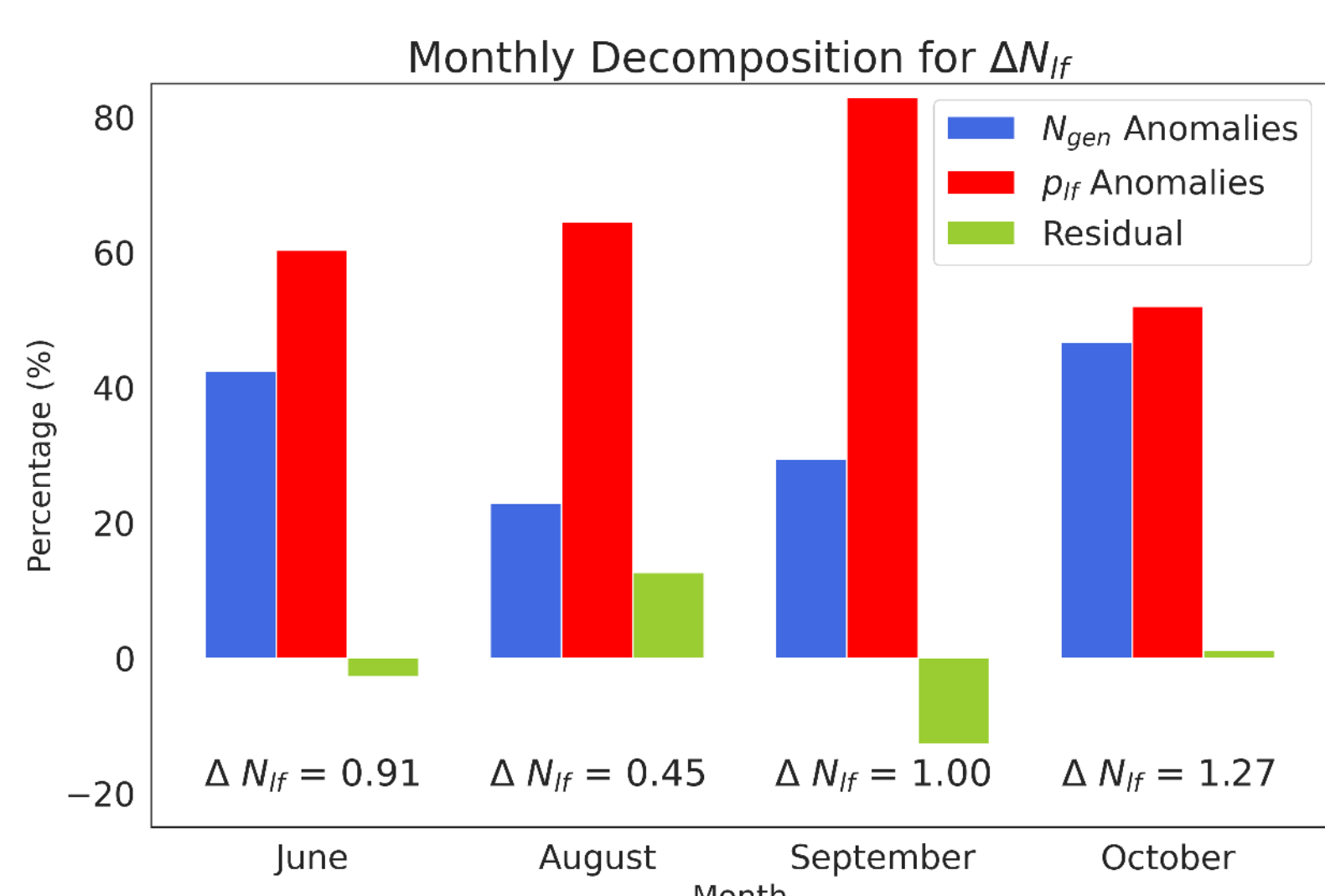


Fig 2. Contributions to the differences in the total number of landfalls between high and low landfall years

Results 1: The ENP has a bimodal distribution of landfalls

- On average, **high landfall** years have **less genesis**.
- The **differences in landfalls** are enhanced for **4 months**: June. August, September and October.

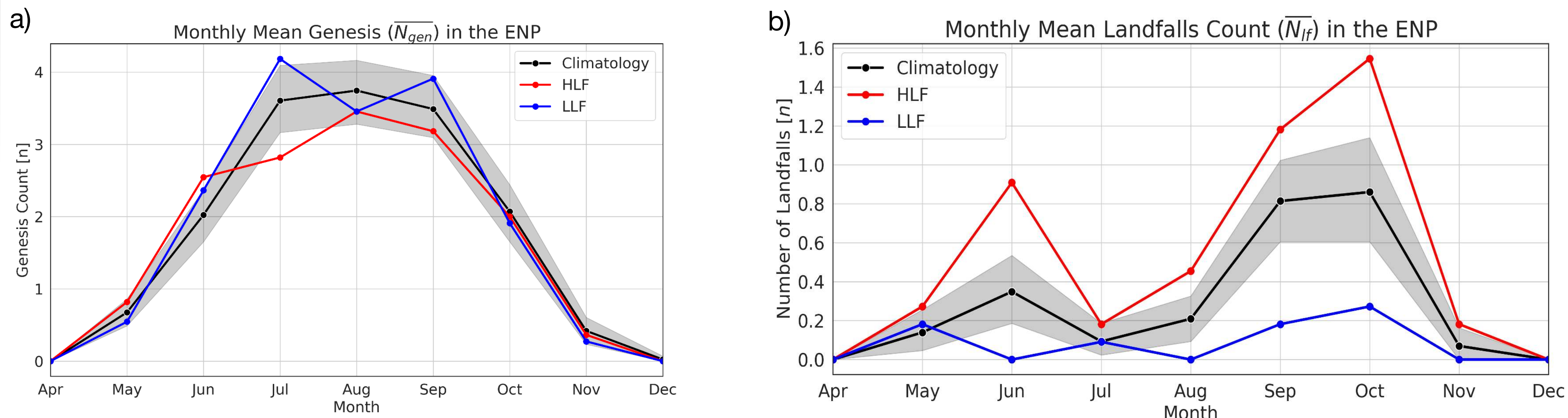


Fig 3. Monthly mean genesis, and b) monthly mean landfalls for climatology (black with 95% CI), HLF (red), and LLF (Green).

Results 2: Significant shift in landfall patterns between HLF and LLF years

- During **High landfall** years, **storms affect** the highly **vulnerable Southwest Mexican coast**.

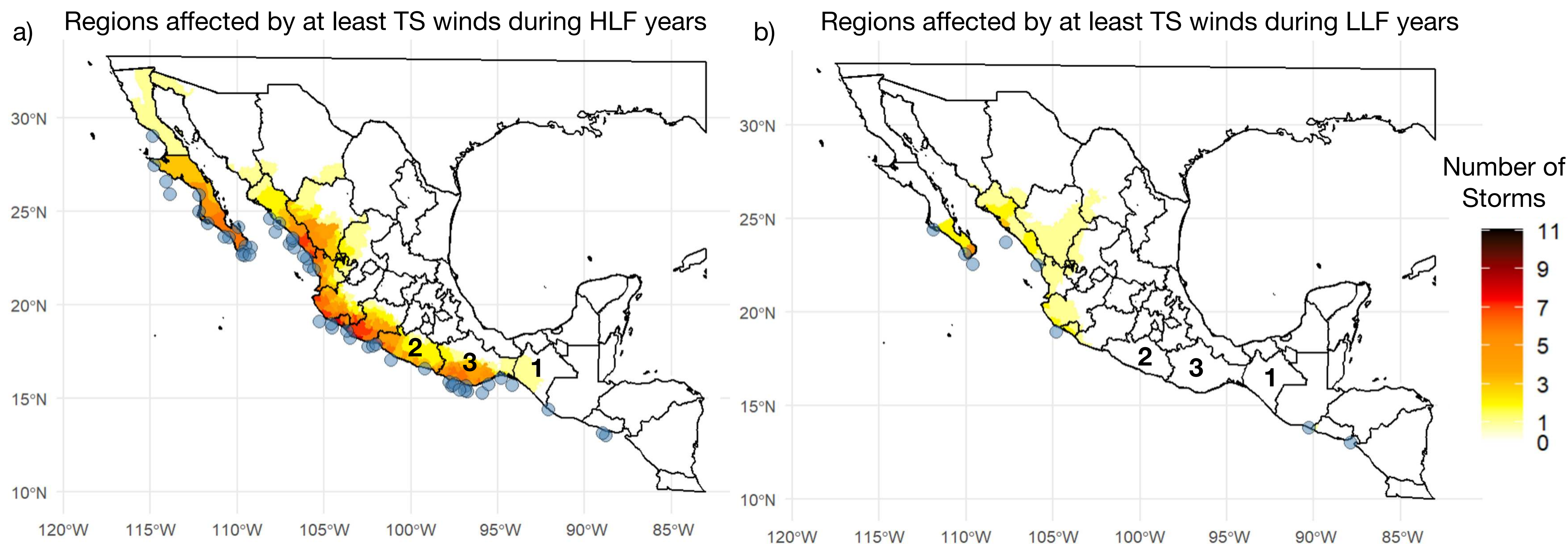


Fig 4. Regions affected by maximum sustained winds of at least TS category (34 knots) for all storms for high (a) and low (b) landfall years. The storm winds were derived using IBTrACS and the “stormwindmodel” (Brooke Anderson, CSU). The blue circles show the landfall location. The numbers highlight Mexico's top 3 states with the highest vulnerability according to the CONEVAL (2020) vulnerability index.

Results 3: Stronger monsoonal winds mark high landfall years

- Most of the **landfall differences** are from TCs forming within **10-20°N**.
- The **regions** with the highest **landfall contribution** highly **correlate** with the stronger **wind reversals**.
- The **bimodal distribution of landfalls** is a product of the **region's orography**.

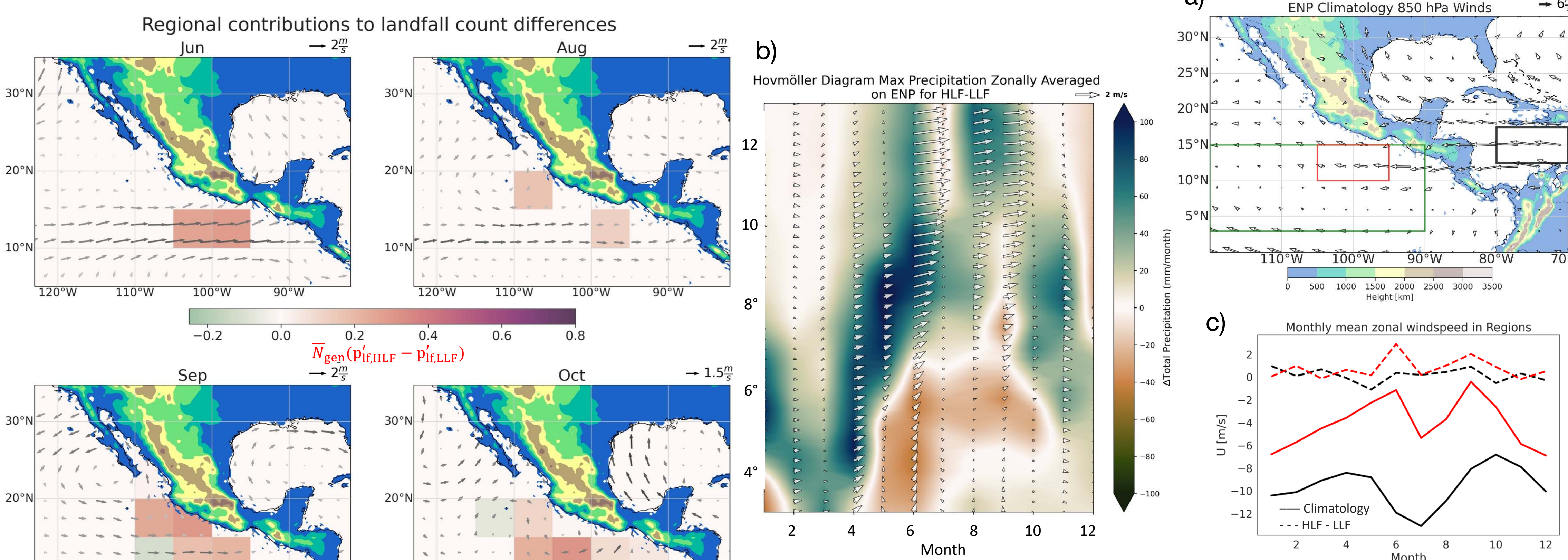


Fig 6. a) Shadings show the orography of the region; the three boxes show where figures b) (green box) and c) (red and black boxes) are calculated. b) shows a Hovmöller diagram for the differences in total precipitation between HLF and LLF years. C) shows a Hovmöller diagram for the monthly mean zonal windspeed in the regions; the thick line is for the climatology, while the dashed line is for the differences between HLF and LLF years.

Future Work: Precipitation analysis and predictability

- Use daily gauge data (MSWEP) to assess the climate risk during HLF and LLF years.
- Perform a predictability analysis using climate variability indices (ENSO, PDO) to improve forecasting in the region. Some preliminary results: feel free to ask!

Manuscript in prep. Comments and suggestions are highly appreciated!